Homework 2
Elementary Linear Algebra
(MATH 2250-03), Fall, 2018

Check: 02/15/2019 Due: 02/18/2019

Section 1.4 - Matrix Equations

1. Let $A = \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Determine the subset $\mathcal{B}$ of vectors in $\mathbb{R}^2$ such that the equation $Ax = b$ has a solution if and only if $b \in \mathcal{B}$. Then, give a set of vectors which span $\mathcal{B}$.

2. Let $v_1 = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ -3 \\ 8 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix}$. Does $\{v_1, v_2, v_3\}$ span $\mathbb{R}^3$? Why or why not?

3. Let $A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix}$. Determine if the columns of A span $\mathbb{R}^4$.

Section 1.5 - Solution Sets of Linear Systems

4. Find a parametric equation of the line M in $\mathbb{R}^2$ which passes through p and q, where
$$p = \begin{bmatrix} 2 \\ -5 \end{bmatrix} \text{ and } q = \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$
(Hint: M is parallel to the vector $q - p$.)

5. Let A be an $n \times n$ matrix, let $b \in \mathbb{R}^n$, and suppose $Ax = b$ has a solution. Explain why this solution is unique precisely when the equation $Ax = 0$ has only the trivial solution $x = 0$.

6. Suppose A is a 3×3 matrix and $y \in \mathbb{R}^3$ is a vector such that the equation $Ax = y$ does not have a solution. Does there exist a vector $z \in \mathbb{R}^3$ such that the equation $Ax = z$ has a unique solution? Why or why not?

Section 1.7 - Linear Independence

7. Find the value(s) of h for which the vectors $v_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} -5 \\ 7 \\ 8 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 1 \\ h \end{bmatrix}$ are linearly dependent in $\mathbb{R}^3$.

8. Let $A = \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & 3 \\ 9 & -3 & 3 \end{bmatrix}$. Solve $Ax = 0$ without performing any row operations. Explain how you obtained this answer and justify your reasoning.

9. Suppose A is an $m \times n$ matrix such that the equation $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$. Must the columns of A be linearly independent? Why or why not?